

Activity - 02

1. STACK AND QUEUE OPERATION :

Stack :

- $push(54) \rightarrow [54]$
- $push(52) \rightarrow [54, 52]$
- $pop() \rightarrow 52$
- $push(55) \rightarrow [54, 55]$
- $push(62) \rightarrow [54, 55, 62]$
- $s = pop() = 62$

QUEUE :

- $enqueue(21) \rightarrow [21]$
- $enqueue(24) \rightarrow [21, 24]$
- $dequeue() \rightarrow \text{Remove } 21$
- $enqueue(28) \rightarrow [24, 28]$
- $enqueue(32) \rightarrow [24, 28, 32]$
- $q = dequeue() = 24$

$0, q = 24$

$s + q = 62 + 24 = 86$

$105 + 60$

$6/*$

$8 -$

$$\begin{array}{|c|} \hline 10 \\ \hline 5 \\ \hline \end{array} + \begin{array}{|c|} \hline 15 \\ \hline \end{array} = \begin{array}{|c|} \hline 60 \\ \hline 6 \\ \hline \end{array} = \begin{array}{|c|} \hline 10 \\ \hline \end{array} = \begin{array}{|c|} \hline 150 \\ \hline 8 \\ \hline \end{array} = \begin{array}{|c|} \hline 142 \\ \hline \end{array}$$

3. minimum number of Enqueue operation

$Q_1 \rightarrow 4$ Elements

$Q_2 \rightarrow 4$ Elements.

4. REVERSE OPERATION

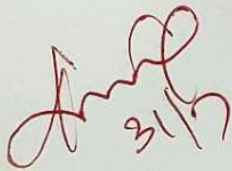
Stack order $O(n)$

5. Postfix Expression

$8 \ 2 \ 3^7 \mid 2 \ 3 \ * \ + \ 5 \mid \ * \ -$

After first $*$ is evaluated the top two stack elements are 6, 5

6. $abc^* + def^7*$ ~~Infix~~ to postfix.


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